CIP-B101 | Basic Computer Skills for Digital Forensics

International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

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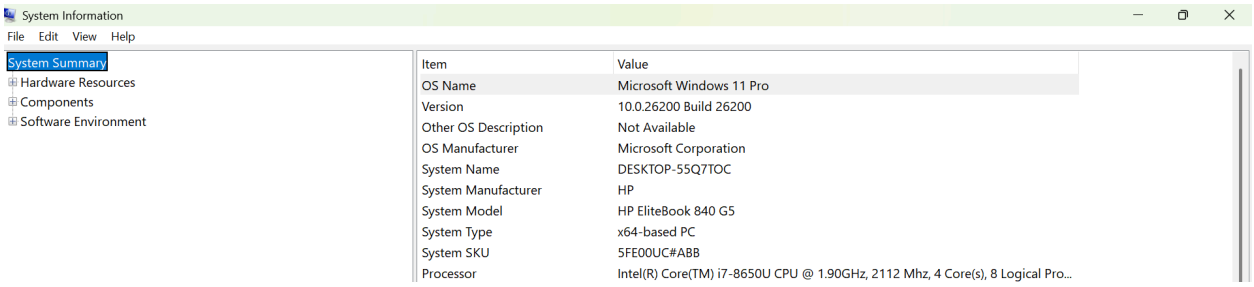
Platform Used: Windows OS and Kali Linux VM

INTRODUCTION:

This lab will help me understand PC system, how to identify key informations, from memory, disk sectors, and partition tables and anlayze these information. It will also help identify system information, map different types of digital evidence and where the evidences can be found.

PART A - SYSTEM IDENTIFICATION AND EVIDENCE LAYER MAPPING

A1. CAPTURE BASIC SYSTEM SPECIFICATION




```
C:\Users\USER>systeminfo

Host Name:                 DESKTOP-55Q7TOC
OS Name:                   Microsoft Windows 11 Pro
OS Version:                10.0.26200 N/A Build 26200
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          USER
Registered Organization:
Product ID:                00330-51791-47546-AAOEM
Original Install Date:     09/10/2025, 09:29:28
System Boot Time:          03/11/2026, 12:46:29
System Manufacturer:       HP
System Model:              HP EliteBook 840 G5
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 142 Stepping 10 GenuineIntel ~1910 Mhz
BIOS Version:              HP Q78 Ver. 01.06.00, 03/01/2019
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
```

A2. CAPTURE STORAGE INFORMATION

```
Administrator: Windows PowerShell

PS C:\WINDOWS\system32> get-disk

Number Friendly Name Serial Number                  HealthStatus         OperationalStatus      Total Size Partition
                                                                                                            Style
------ ------------- -------------                  ------------         -----------------      ---------- ----------
0      SAMSUNG MZ... 0025_3881_91B0_78F0.           Healthy              Online                  476.94 GB GPT


PS C:\WINDOWS\system32> get-partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_samsung_mzvlb512#5&84ce291&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    100 MB System
2                            105906176                                   16 MB Reserved
3                C           122683392                                475.72 GB Basic
4                            510921801728                               1.1 GB Recovery


PS C:\WINDOWS\system32> get-volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         NTFS           Fixed     Healthy      OK                    114.8 MB     1.1 GB
                         FAT32          Fixed     Healthy      OK                   29.33 MB      96 MB
C                        NTFS           Fixed     Healthy      OK                  186.32 GB 475.72 GB
```

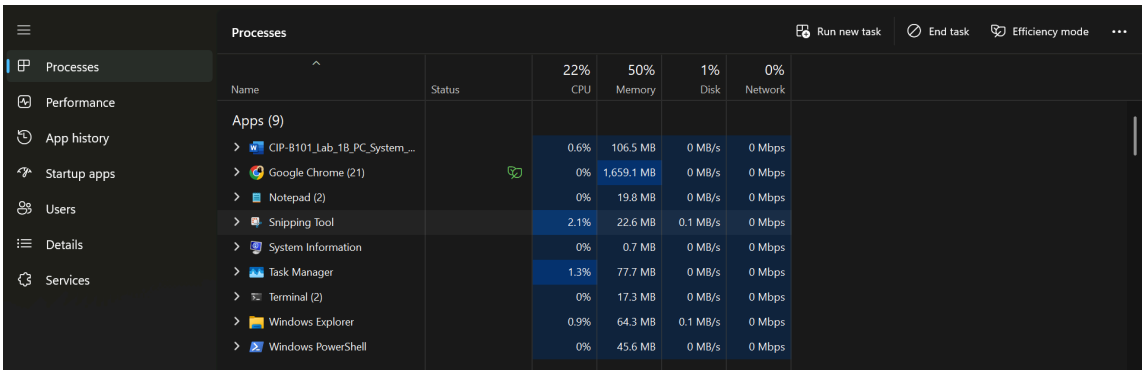
A3. MAP EVIDENCE SOURCES TO SYSTEM LAYERS

Layer/Source	**Evicence**	**Where it may be found**	**Why it matters**
User layer	Login and Log out, file changes, file modification,		It helps examiner connect user to system activities
Operating System	System configuration changes, scheduled tasks, user event.	Event viewer, device manager, windows logs	Can reveal attempted unauthorized access., May reveal changes to system settings, devices, and configurations.
Application Layer	Browser & download history, search history, application usage, Recently opened application files.	Browsers , cache, saved cookies, etc.	It helps investigators understand what a user did within specific applications, including browsing, communicating, downloading files, editing documents, and accessing online services.
Network Layer	Remote access activity, actie network, connections	VPN logs, RDP logs and firewall records	It helps investigators trace communications, identify suspicious connections,

			investigate possible data exfiltration, and establish whether a computer communicated with potentially malicious systems.
Memory/Process Layer	Running processes, PIDs, runing services, resource consumption	Task Manager, PowerShell, process inspection tools	It helps investigators examine the state of a computer at a particular moment, identify suspicious processes.

PART B - PROGRAM, PROCESS AND THREAD INVESTIGATION

B1. OBSERVE RUNNING PROCESSES



B2. EXPLAIN THE DIFFERENCE

- In your report, explain the difference between a program, a process, and a thread in your own words.

  All three represent layers of execution on a computer, A program is a set of instructions that is waiting to be executed by a computer. While a process is an active program that is

currently being executed. A thread is like different processes broken down into smaller executable chunks.

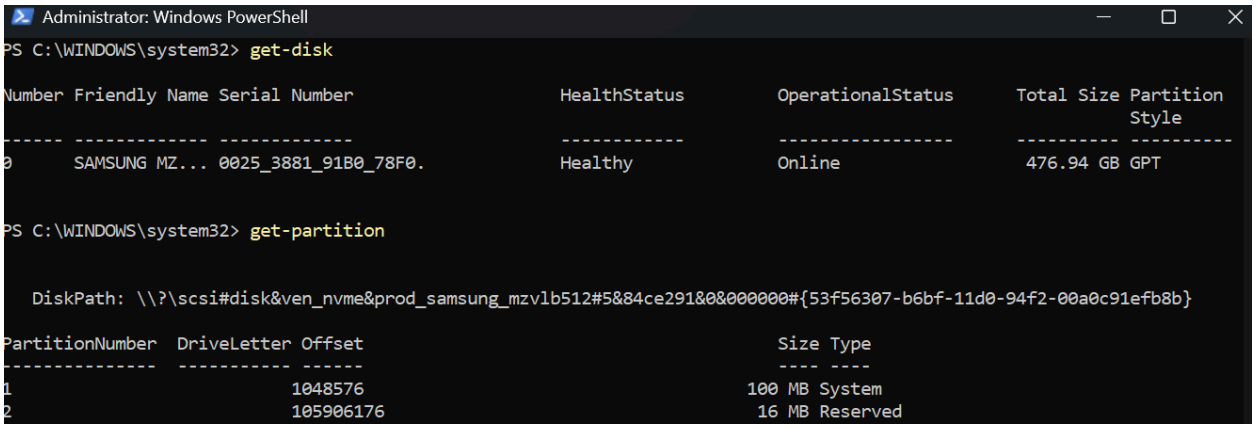
- Explain why process information may be useful during malware analysis or live forensic triage.

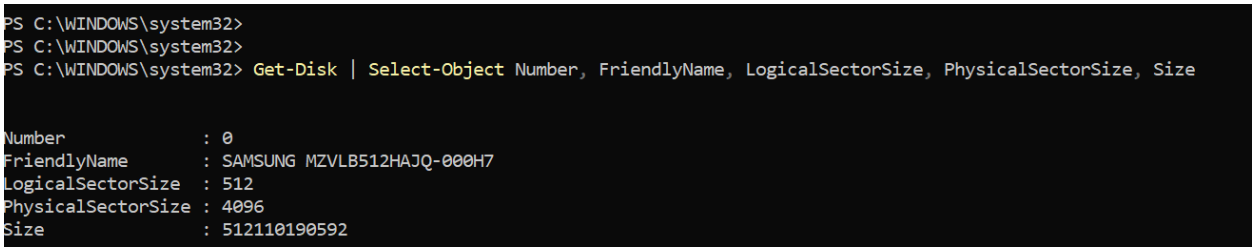
  Analyzing active processes is a critical component of live forensic triage because a process represents a program in its active execution state. In a live environment, such as viewing Windows Task Manager, the operating system stores running processes alongside their real-time resource consumption, including CPU, memory, disk, and network utilization. When a system is compromised by malware, volatile process information serves as a primary indicator of compromise (IoC) in two ways: **Anomaly Detection:** Unauthorized or malicious executables running in memory can be immediately identified, isolated, and flagged for further analysis. **Resource Profiling:** Malicious activity, such as data exfiltration, or background scanning, frequently triggers abnormal spikes in system resources. Monitoring process metrics allows investigators to pinpoint exactly which executable is generating the abnormal network traffic or CPU load.

PART C - DISK SECTOR AND DISK SIZE CALCULATION

Command used:

- Get-Disk
- Get-Disk | Select-Object Number, FriendlyName, LogicalSectorSize, PhysicalSectorSize, Size
- Get-Partition





Partition 1:

Fomular:

Start sector = offest / logical sectorsize

Sector count = start sector* *512/1024*1024*1024

Start sector = 1048576/512 = 2048

Disk size = 2048*512/1024*1024*1024

= 1,048,576/1024*1024*1024

= 0.0009765625GB

Partition 2

Start sector = 105906176/512 = 206,848

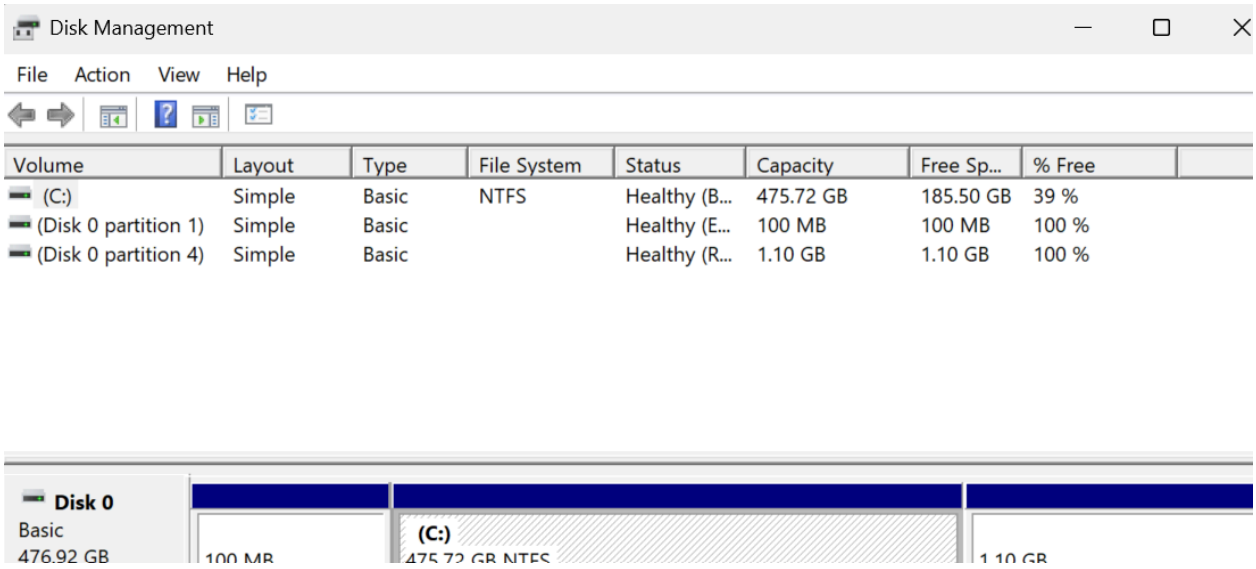
Disk size = 206,848*512/1024*1024*1024

= 105,906,176/1024*1024*1024

= 0.0986GB

PART D - SAFE PARTITION TABLE INSPECTION

D1. Windows method - Disk Management, PowerShell and DiskPart



PART E - MBR, GPT AND BOOT PROCESS EXPLANATION

E1: EXPLAIN THE BOOT PROCESS

A. How PC Boots:

The boot process is the sequence of events that takes place when a computer is switched on, from receiving power to loading the operating system and making the computer ready for use.

1. Power-On: The boot process begins when the power button is pressed. The power supply unit (PSU) supplies electrical power to the computer's components, including the motherboard, CPU, RAM, and storage devices. The CPU receives power and begins executing instructions from the firmware stored on the motherboard.

2. CPU Initialization: The CPU starts executing its initial instructions from a predefined memory location, directing it to the BIOS or UEFI firmware. This firmware contains the instructions required to initialize the hardware and begin the boot process.
3. POST (Power-On Self-Test): The BIOS/UEFI performs POST to check whether essential hardware components are functioning correctly. It checks components such as RAM, CPU, keyboard, and other connected hardware. If an issue is detected, the computer may display an error message, produce beep codes, or stop the boot process.
4. BIOS/UEFI Initialization: After POST, the BIOS or UEFI firmware initializes and identifies connected hardware devices, including storage drives, USB devices, and other peripherals. It prepares the hardware required for the operating system to load.

- BIOS (Basic Input/Output System): The traditional firmware interface used to initialize hardware and start the boot process.
- UEFI (Unified Extensible Firmware Interface): The modern replacement for traditional BIOS, offering additional features such as Secure Boot, support for GPT disks, and an EFI System Partition.

5. Boot Device Selection: Based on the configured boot order, the BIOS/UEFI searches for a bootable device, such as an SSD, HDD, or USB drive. Once a suitable boot device is identified, the firmware proceeds to locate the appropriate boot information.
6. MBR/GPT or EFI Boot Records: The firmware uses the disk's partitioning structure to locate the boot information.

- MBR (Master Boot Record): Used in traditional BIOS booting. It is located in the first sector of the disk and contains boot code and the partition table.
- GPT (GUID Partition Table): Commonly used with UEFI systems. It contains partition information, including partition entries and backup partition metadata.
- EFI System Partition (ESP): On UEFI systems, this is a dedicated FAT-formatted partition containing EFI bootloader files that the firmware can execute.

7. Boot Loader/Boot Manager Execution: Once the boot information is located, the firmware transfers control to the bootloader or boot manager.

- In BIOS/MBR systems, the boot code in the MBR locates and transfers control to the next stage of the bootloader.

- In UEFI/GPT systems, the firmware loads an EFI executable from the EFI System Partition. For example, Windows uses Windows Boot Manager (bootmgfw.efi). The boot manager identifies the operating system to load and starts the appropriate operating system loader.

8. Operating System Loader: The operating system loader prepares the operating system for execution. For example, Windows uses winload.efi on UEFI systems.
9. Kernel Loading: The operating system kernel is loaded into RAM and begins executing. The kernel is the central component of the operating system responsible for managing system resources, memory, processes, devices, and communication between hardware and software.
10. Kernel Initialization: The kernel initializes essential system components, loads additional device drivers, establishes memory management, and prepares the environment required for system services and applications to run.
11. System Services and Processes Start: The operating system starts essential background services and system processes.
12. User Login and Operating System Control: The login interface is presented to the user. Once authentication is successful, the desktop environment loads, allowing the user to interact with applications and system resources. At this stage, the operating system has full control of the computer's hardware and software resources.

B. State whether your own system uses BIOS/Legacy or UEFI based on system information evidence. **Answer: My PC uses UEFI**

PART F - FILE SYSTEM IDENTIFICATION AND INTERPRETATION

- Identify the file system used by at least two volumes or partitions on your system.

```
S C:\WINDOWS\system32> get-volume

riveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
---------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                        NTFS           Fixed     Healthy      OK                    114.8 MB    1.1 GB
                        FAT32          Fixed     Healthy      OK                    29.33 MB     96 MB
                        NTFS           Fixed     Healthy      OK                   186.32 GB 475.72 GB
```

- For Linux systems, identify whether partitions use ext4, XFS, btrfs, swap, or another file system. **Answer: ext4**

```
┌──(kali㉿kali)-[~/jesudamilola_cip-b101-lab1a]
└─$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT

NAME    SIZE TYPE FSTYPE MOUNTPOINT
sda    80.1G disk
└─sda1 80.1G part ext4   /
```

- Explain why a file system is important in digital forensics.

Part G - Final Practical Assignment

Main assignment: Verify the size of partitions 1 and 2 of a PC using a partition table. Record the verification process step-by-step. Calculate the size of each partition in GB. No full credit will be awarded if you do not use partition table evidence.

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> get-disk

Number Friendly Name Serial Number                  HealthStatus         OperationalStatus      Total Size Partition
                                                                                                            Style
------ ------------- -------------                  ------------         -----------------      ---------- ----------
0      SAMSUNG MZ... 0025_3881_91B0_78F0.           Healthy              Online                  476.94 GB GPT


PS C:\WINDOWS\system32> get-partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_samsung_mzvlb512#5&84ce291&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                   100 MB System
2                            105906176                                  16 MB Reserved
```

Task	Required Evidence	Student notes/Answer

Identify Partition 1	Screenshot of partition table and command output	https://drive.google.com/file/d/1Ww7rCUOAC4xbXe0loBW5P5lxR_iYUPfF/view?usp=sharing
Record partition 1 sector count	Start sector, end sector or sector count	2048
Calculate partition 1 size	Formula and result in GB	0.0009765625GB
Identify partition 2	Screenshot of partition table and command output	https://drive.google.com/file/d/1Ww7rCUOAC4xbXe0loBW5P5lxR_iYUPfF/view?usp=sharing
Record partition 2 sector count	Start sector, end sector or sector count	206,848
Calculate partition 2 size	Formula and result in GB	0.986GB
Compare with OS reported size	Disk Management, Get-Volume, lsblk, or df output	Partition 1 size is 100mb and the disk calculation to GB is = 0.0009765625GB
Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	Windows display values in round numbers, but the calculation gives a much more precise number.

8. STUDENT REPORT TEMPLATE

- File system identification: explain the file systems found.

  1. FAT: Windows file allocation table, it uses a table to keep track of where files are stored on a disk. Hence, it tracks file storage through an allocation table.

2. NTFS: Tracks files and their metadata through MFT records.
3. EXT4: linux file system, it tracks file metadata through inode structures.

SUMMARY:

This course module provide the foundation for understanding how a computer operates, manages hardware and software resources, executes processes, stores and organizes data, and generates system activities. Understanding system information, running processes, system logs, disk structures, partitioning schemes (MBR/GPT), boot processes, and file systems (FAT, NTFS, EXT4) helps digital forensic investigators identify relevant evidence, understand where it is stored, detect potentially malicious activities, reconstruct system events, and effectively examine and preserve digital evidence.

APENDIX:

